

The Gradient Descent Algorithm ⚙️

Gradient descent is an iterative process that repeatedly updates the model parameters w and b to minimize the cost function $J(w, b)$.

The core of the algorithm is the following update rule, which is repeated until the model converges:

$$\begin{aligned} w &:= w - \alpha \frac{\partial}{\partial w} J(w, b) \\ b &:= b - \alpha \frac{\partial}{\partial b} J(w, b) \end{aligned}$$

Components of the Update Rule 🤖

Let's break down what each part of the rule means:

- **$:=$ (The Assignment Operator):** In this context, the equal sign means we are **assigning** a new value to the parameter. We calculate the entire expression on the right and use it to update the value of w or b on the left.
 - **α (Alpha, The Learning Rate):**
 - Alpha is a small positive number (e.g., 0.01) that controls **how big of a step** you take downhill on each iteration.
 - A large α means taking aggressive, large steps.
 - A small α means taking small, cautious baby steps.
 - Choosing a good value for α is a critical part of making the algorithm work well.
 - **$\frac{\partial}{\partial w} J(w, b)$ (The Derivative Term):**
 - You don't need to know calculus to understand its role here.
 - This term calculates the **direction of the steepest descent**. It tells the algorithm *which way* to go to reduce the cost most effectively.
 - Multiplying the direction (derivative) by the step size (α) determines the overall update for the parameter.
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Correct Implementation: Simultaneous Updates ✅

A crucial detail for implementing gradient descent correctly is to perform **simultaneous updates** for all parameters. This means you must calculate the new values for w and b *before* updating either of them.

- **Correct Method:**
 1. Calculate the update for w and store it in a temporary variable: `temp_w = w - ...`

2. Calculate the update for b and store it in another temporary variable: $\text{temp_b} = b -$
...
3. Update w and b with the values from the temporary variables: $w = \text{temp_w}, b = \text{temp_b}$.

- **Incorrect Method (Sequential Update) ❌:**

1. Calculate the new w and immediately update it.
2. Use this **newly updated** w to calculate the update for b .
 - This is technically a different algorithm and is not the standard implementation of gradient descent.